

Scalable Quantum-Enhanced Distributed Order Management with Pauli Correlation Encoding

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Team: The Quantum DOMinators

The Fundamental Challenge: A Combinatorial Explosion

- **Problem:** Need to find the optimal mix of routing decisions across a network of Distribution Centers (DCs) when default locations face stockouts or capacity limits.
- **Goal:** To maximize expected fulfilled revenue while minimizing financial risk (shortfall penalties and freight costs).
- **The Issue:** The exponential growth in the number of possible routing configurations (D^N) makes it impossible for classical computers to check every combination, a problem known as a combinatorial explosion.

Defining the Problem: Master-Recourse Model with Constraints

- We formulate DOM using a hybrid architecture to ensure operational feasibility.
- **Master Routing (Quantum):** A binary decision variable $y_o \in \{0, 1\}$ dictates if order o stays at its default DC or diverts.
- **Operational Recourse (Classical):** A fast MILP calculates the exact integer fulfillment and enforces strict linear inequality constraints (inventory, dock, labor).

The Unified Objective:

$$Z(y) = \sum_{o,s} p_{os} x_{os} - \rho \sum_{o,s} p_{os} u_{os} - \sum_o c_{o,d(o;y)} v_o$$

Pauli Correlation Encoding: A Resource-Efficient Approach

- **What it is:** Pauli Correlation Encoding (PCE) is a novel and efficient method for quantum optimization. It maps classical binary variables to multi-body Pauli-matrix correlations.
- **Qubit Efficiency:** PCE drastically reduces the number of qubits needed for optimization problems. This is crucial for near-term quantum devices.
- **Our Extension:** We adapted PCE's application to a hybrid DOM framework, combining quantum combinatorial search with exact classical recourse constraints.

PCE: The Encoding Rule and Practical Implementation

Encoding rule: Each classical binary variable y_i is encoded as the sign of the expectation value of a unique multi-qubit Pauli string (correlator) as Π_i follows:

$$y_i = (1 - \text{sign}\langle\psi(\theta)|\Pi_i|\psi(\theta)\rangle)/2$$

Here, $|\psi(\theta)\rangle$ is a parameterized quantum state. In this work, we use the hardware-efficient `efficient_su2` ansatz to construct $|\psi(\theta)\rangle$.

In Practice: In a real-world application, the sharp $\text{sign}(\ast)$ function is replaced by a smoother hyperbolic tangent function $\tanh(\alpha\ast)$, which is better suited for optimization. We iteratively grow α to harden the decisions.

Scaling: n -qubit, k -body correlators can solve problems of size $m = O(n^k)$.

PCE Implementation: 8 Variables on 3 Qubits

2-body correlators follow the following scaling:

Group 1 (X)		Group 2 (Y)		Group 3 (Z)	
Π_1	IXX	Π_4	IYY	Π_7	IZZ
Π_2	XIX	Π_5	YIY	Π_8	ZIZ
Π_3	XXI	Π_6	YYI	Π_9	ZZI (unused)

3 qubits can encode 9 binary variables

4 qubits can encode 18 binary variables

5 qubits can encode 30 binary variables

n qubits can encode $3 \times \frac{n(n-1)}{2}$ binary variables

Hybrid Variational Optimization of PCE encoded DOM Objective

- **Optimizer:** COBYLA, driven by an Augmented Lagrangian loss function to handle capacity constraints.
- **Simulation type:** We use both Ideal (State Vector) and Noisy simulations with a real backend snapshot (FakeManilaV2), running 4096 shots.
- This approach prepares algorithms for real-world conditions, testing their robustness against errors on near-term devices.
- This ensures solution performance reflects what is achievable on current quantum hardware, even before direct hardware access.

Capacity-Aware Training Loss: ALM & Hinge Loss

- **The Challenge:** The quantum optimizer (COBYLA) must respect strict classical capacity constraints (dock, case-pick, pallet-pick) during training.
- **Our Solution:** We incorporate inequality constraints into the objective function using a hinge loss formulation, updated via an Augmented Lagrangian variant.

The Loss Function:

$$\mathcal{L}(\theta) = -\frac{V(\tilde{y}(\theta))}{V_{scale}} + \sum_k \left[\lambda_k^{(t)} g_k^+ + \frac{\mu_k}{2} (g_k^+)^2 \right]$$

- g_k^+ represents the normalized constraint violation $\max(0, g_k)$.
- $\mu_k = 10.0$ acts as a fixed quadratic penalty weight to maintain a smooth landscape.
- λ_k (Lagrange multipliers) are iteratively updated to enforce strict feasibility.

Results & Analysis: Solving a Distributed Order Management Problem

Optimal Routing: Our quantum solver returned a bitstring representing an optimal assignment for the 8 contested orders across the supply chain network.

Benchmark Comparison:

- We compared our result against an exhaustive exact solver, which found an optimal value of \$1,080,634.47 with ideal simulation.
- The final output from the Aer simulator with noise from a real backend (FakeManilaV2) run identified the identical routing configuration, achieving the exact same cost value of \$1,080,634.47 after a 1-bit classical repair.

Performance: Our quantum-derived solution is as good as the benchmark, proving the effectiveness of our PCE-MILP approach on real hardware (delivering a 3.17% financial lift).

THANK YOU